

Replacing Micron 28F128M29EW with Macronix MX29GL128F

1. Introduction

Macronix offers MX29GL_F high performance parallel flash in densities from 128Mb to 1Gb. MX29GL_F and Micron® M29EW devices have similar features, performance, and command codes, but there are some differences as indicated in blue text below. This application note explains how to accommodate those differences when migrating from Micron® 28F128M29EW devices to equivalent Macronix MX29GL128F parallel flash products.

The information in this document is based on datasheets listed in Section 11. Newer versions of the datasheets may override the contents of this document.

2. Features

Both flash device families have similar features and functions as shown in Table 2-1.

Table 2-1: Feature Comparison

Type / Function	Macronix MX29GL128F	Micron 28F128M29EW
VCC Voltage Range	2.7V ~ 3.6V	2.7V ~ 3.6V
I/O Voltage Range	2.7V ~ 3.6V (H/L type ^{*1}) 1.65V ~ 3.6V (U/D type ^{*2})	1.65V ~ 3.6V
Bus Width	x16 / x8	x16 / x8
Sector Size	128KB	128KB
Page Read Buffer Size	8Words / 16Bytes	8Words / 16Bytes
Write Buffer Size	32Words / 64Bytes	256Words / 256Bytes ^{*4}
WP# pin function	Highest/Lowest address sector	Highest/Lowest address sector
Software Protected Mode	Password (64bits) Solid Protection ^{*3}	Password (64bits) Non-volatile Protection ^{*3}
OTP Security Region	128Words / 256Bytes	128Words / 256Bytes
Blank Check Command ^{*5}	-	Yes
Manufacture ID	C2h	89h
Device ID	227E/2221/2201	227E/2221/2201
Package	56-TSOP (14x20mm) 64-LFBGA (11x13mm)	56-TSOP (14x20mm) 64-LFBGA (11x13mm)

Note:

1. Macronix offers 'H/L' type
H type is "VI/O = Vcc = 2.7 ~ 3.6V, highest address sector protected."
L type is "VI/O = Vcc = 2.7 ~ 3.6V, lowest address sector protected."
2. Macronix 'U/D' is same as Micron® default part number.
U type / Micron® device is "VI/O = 1.65 ~ Vcc, Vcc = 2.7 ~ 3.6V, highest address sector protected."

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- D type / Micron[®] device is "VI/O = 1.65 ~ Vcc, Vcc = 2.7 ~ 3.6V, lowest address sector protected."*
3. Solid Protection is same function as Non-volatile Protection, which is just different naming
 4. Micron[®] write buffer is 256-Bytes in x8 mode and 256-Words in x16 mode.
 5. Micron[®] provides Blank Check command to determine memory cell that are programmed or over-erase.

3. Performance

Table 3-1 shows MX29GL128F series and 28F128M29EW series Read/Write performance.

Table 3-1: Read Function Performance (Random Read and Page Read)

Read function	Macronix MX29GL128F	Micron 28F128M29EW
Random Read Access time (Taa/Tce or tAVQV/tELQV)	90ns (H/L type) 110ns (U/D type)	60ns (BGA package) 70ns (TSOP package)
Page Access time ^{*1} (Tpa or tAVQV1)	25ns (H/L type) 30ns (U/D type)	25ns

Note:

1. Random Read Access time: Macronix performance is VIO voltage dependent, and Micron[®] is package dependent.
2. System needs to align read speed for replacement.
3. Align 'wait state' setting of the controller or SoC, if speed is critical factor in system.

Table 3-2: Write Function Performance (Program and Erase)

Write Function		Macronix MX29GL128F	Micron 28F128M29EW
Write Buffer	32 word ^{*1} (typ)	120us	85us
Program time	256 word ^{*1} (typ)	N/A	284us
Word Program Time (typ)		11us	15us
Sector Erase Time	typ	0.6s	0.5s
	max	5s	4s
Chip Erase time (typ)		64s	N/A
Write/Erase Cycles (Endurance)		100,000	100,000

Note:

1. The maximum buffer sizes of the Macronix and Micron devices are different. Using only to 32-words of the buffer and aligning the write buffer addresses on a 32-Word page boundary provides system compatibility.

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3-1. Write Buffer Length Alignment

The Write Buffer sizes are different between the two devices and software modification is necessary to adjust for the difference. Macronix MX29GL12835F Write Buffer size is 32-Words. Micron 28F128M29EW Write Buffer size is 256-Bytes in x8 mode and 256-Words in x16 mode. There are two suggested methods for adapting to the difference in Write Buffer sizes:

- (i) Use a maximum Write Buffer length of 32-Words for both devices.
- (ii) Read the maximum Write Buffer length from the CFI table and adjust the algorithm to use this length. The Write Buffer length parameter is located at offset address 2Ah (word mode) of the CFI table.

Regardless which method is adopted, the Write Buffer addresses need to fall within a 32-Word page boundary when using the Macronix device.

4. DC Characteristics Comparison

Table 4-1 shows that Read and Write currents are similar.

Table 4-1: Read / Write Current

DC Characteristic	Condition	Macronix MX29GL128F	Micron 28F128M29EW
Read Current	typ	20mA @ 5MHz	20mA @ 5MHz
	max	50mA @ 5MHz	25mA @ 5MHz
Page Read Current	typ	2mA @ 10MHz	12mA @ 13MHz
	max	10mA @ 10MHz	16mA @ 13MHz
Standby Current	typ	30uA	50uA
	max	100uA	120uA
Write Current	typ	26mA	35mA
	max	30mA	50mA
Deep Power Down	typ	10uA	N/A

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Generally, both have similar input and output voltage ranges as shown in Table 4-2.

Table 4-2: Input / Output Voltage

DC Characteristic	Condition	Macronix MX29GL128F	Micron 28F128M29EW
Very High Voltage	-	9.5V ~ 10.5V ^{*1}	11.5V ~ 12.5V
Input Low Voltage	min	-0.1V	-0.5V
	max	0.3V _{I/O}	0.8V
Input High Voltage	min	0.7V _{I/O}	2.0V
	max	V _{I/O} +0.3V	V _{I/O} +0.5V
Output Low Voltage	max	0.45V	0.15V _{I/O}
Output High Voltage	min	0.85V _{I/O}	V _{I/O} – 0.2V

Note:

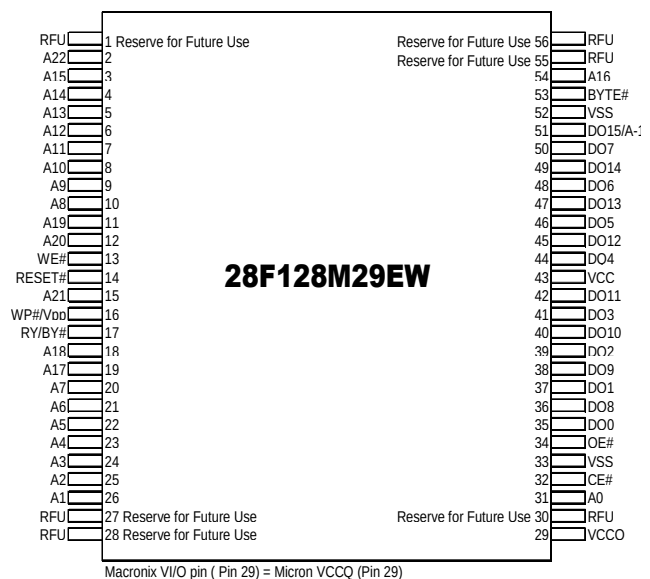
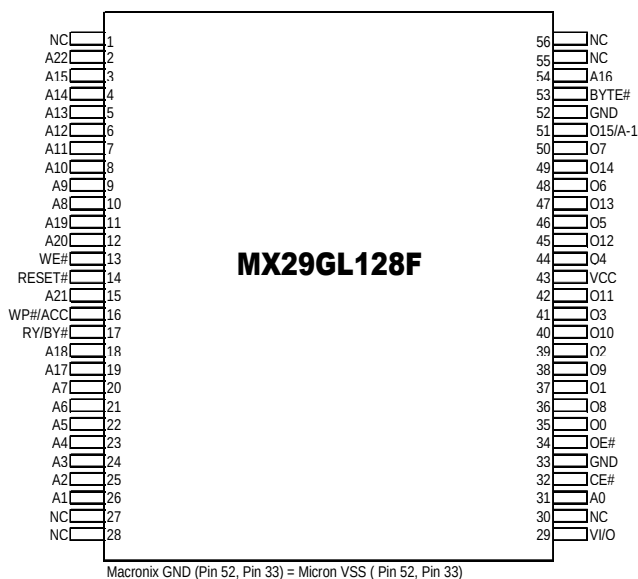
1. The major difference is Very High Voltage range which may damage the flash device if the maximum level is exceeded.

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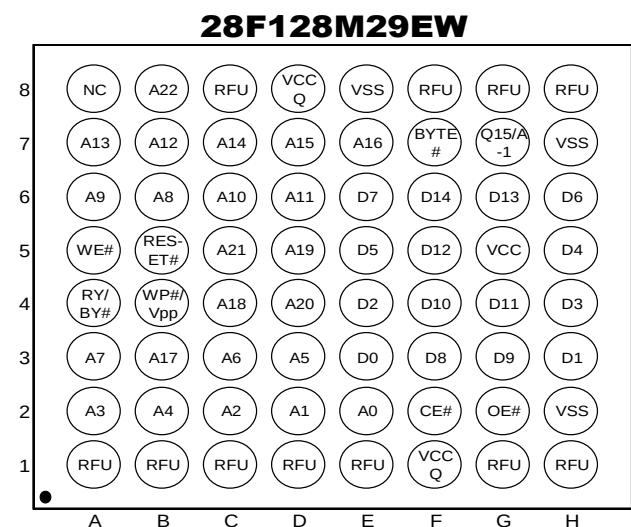
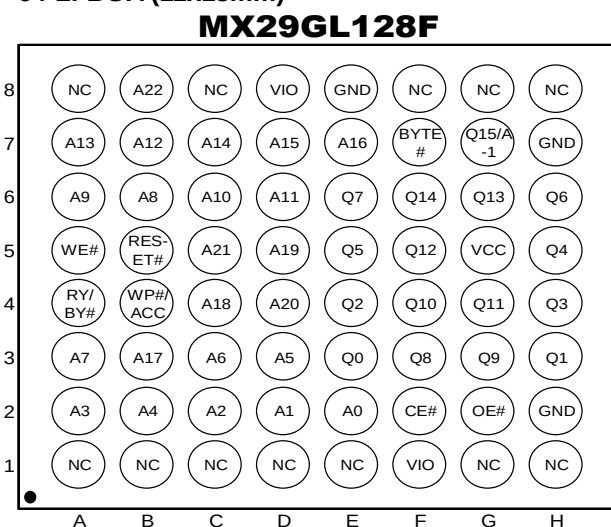
5. Hardware Consideration

The Macronix device has a similar footprint with the Micron® device. Please refer to the 56-TSOP & 64-LFBGA diagrams below.

56-TSOP (14x20mm)



64-LFBGA (11x13mm)



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6. Software Considerations

Basic command sets and write status checking methods are similar between both flash families. The algorithm descriptions may be slightly different, but the concepts are the same.

Micron does offer an additional Blank Check Setup (76h) and Blank Check Read (29h) command (to determine if a specific block is completely erased) which is not supported by Macronix flash.

6-1 Basic Command Table (Word mode)

MX29GL128F and 28F128M29EW have same basic command set. The Read operation and Write command could be used directly without any modification. Table 6-1 shows the command set in Word mode.

Table 6-1: Basic Command Table

Command		Read	Reset	Program	Write to Buffer	Chip Erase	Sector Erase	Program/Erase Suspend	Program/Erase Resume
1 st Bus Cycle	Addr	Addr	XXX	555h	555h	555h	555h	XXX	XXX
	Data	Data	F0h	AAh	AAh	AAh	AAh	B0h	30h
2 nd Bus Cycle	Addr			2AAh	2AAh	2AAh	2AAh		
	Data			55h	55h	55h	55h		
3 rd Bus Cycle	Addr			555h	SA ^{*1}	555h	555h		
	Data			A0h	25h	80h	80h		
4 th Bus Cycle	Addr			Addr	SA	555h	555h		
	Data			Data	N-1 ^{*2}	AAh	AAh		
5 th Bus Cycle	Addr				WA ^{*3}	2AAh	2AAh		
	Data				WD ^{*4}	55h	55h		
6 th Bus Cycle	Addr				WBL ^{*5}	555h	SA		
	Data				WD	10h	30h		

Note:

1. SA: Sector Address
2. N-1: Word Count
3. WA: Write Address
4. WD: Write Data
5. WBL: Write Buffer Location

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6-2 Write Status Checking Method

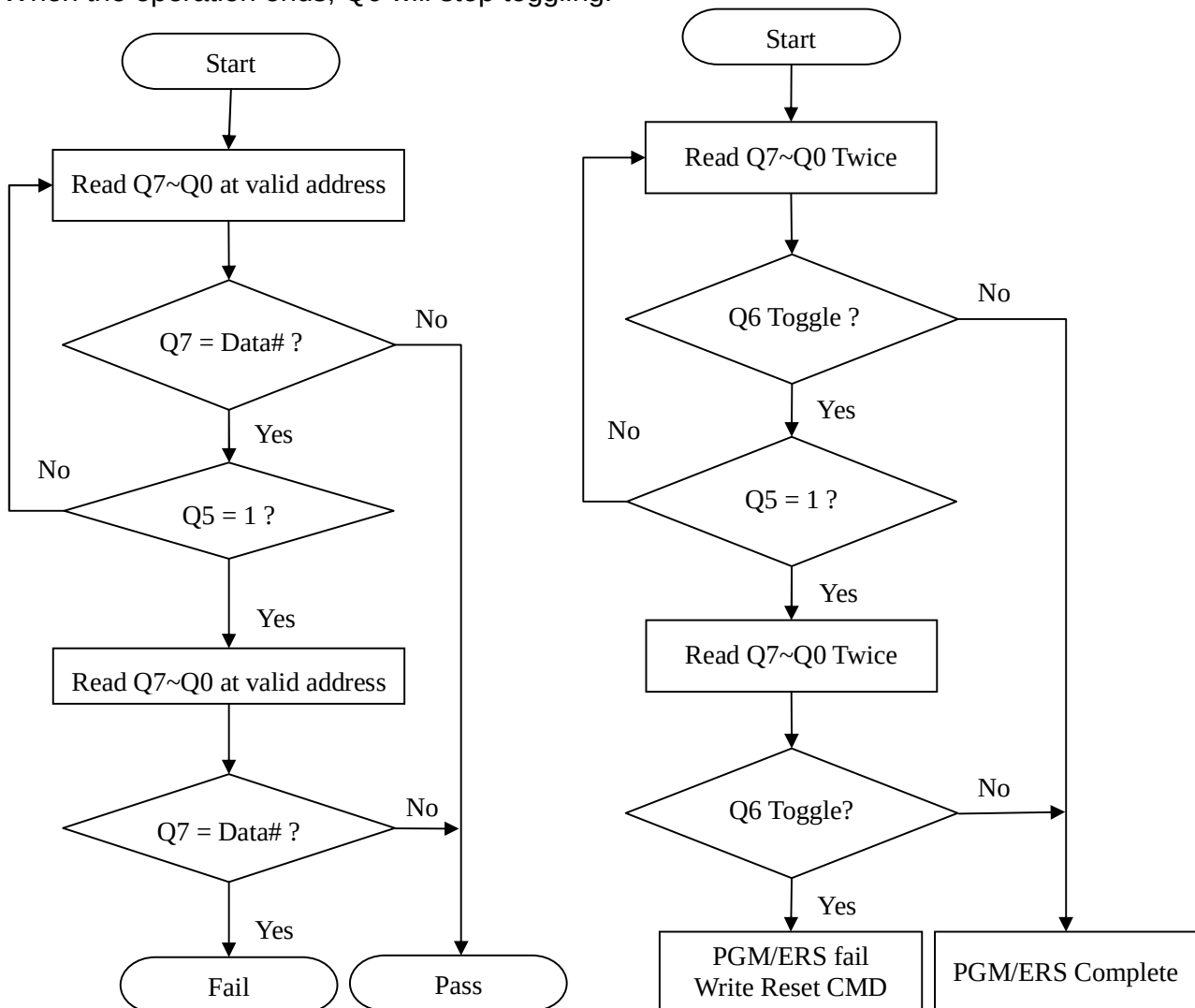
When a flash program/erase operation is in progress, either the "Polling Method" or "Toggle Bit Method" may be used to monitor the operation. Both are standard algorithms in parallel flash and can be used for both device families.

Polling Method:

Polling method checks Q7 (data complement bit) and Q5 (time out bit) values during the program/erase operation. After the operation has finished, Q7 will output true Data.

Toggle Bit Method:

The toggle bit method checks the Q6 (toggle bit) value during the program/erase operation. When the operation ends, Q6 will stop toggling.



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7. Manufacturer ID & Device ID Command

Manufacturer IDs are different and permit software to identify the device manufacturer. The same command set is used read the different Manufacturer IDs. Device IDs are the same.

Table 7-1: Manufacturer and Device ID Command Definitions

Flash Vender		Manufacturer ID		Device ID	
		MX29GL128F	28F128M29EW	MX29GL128F	28F128M29EW
1 st Bus Cycle (command)	Address	555h	555h	555h	555h
	Data	AAh	AAh	AAh	AAh
2 nd Bus Cycle (command)	Address	2AAh	2AAh	2AAh	2AAh
	Data	55h	55h	55h	55h
3 rd Bus Cycle (command)	Address	555h	555h	555h	555h
	Data	90h	90h	90h	90h
4 th Bus Cycle (ID output)	Address	X00h	X00h	X01h	X01h
	Data	C2h	89h	227Eh	227Eh
5 th Bus Cycle (ID output)	Address			X0Eh	X0Eh
	Data			2221h	2221h
6 th Bus Cycle (ID output)	Address			X0Fh	X0Fh
	Data			2201h	2201h

Note.

1. Device ID can be read out after Manufacturer ID with proper address and does not need another command sequence.
2. Use Reset command (F0h) to return to normal read mode.

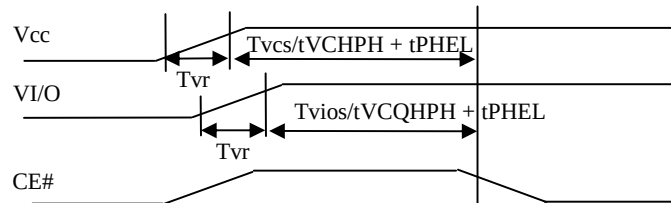
8. Power-Up Timing

Macronix and Micron[®] power on sequences are similar, but the timing is slightly different. Check system timing to determine if a timing adjustment is needed.

Table 8-1: Power-Up Timing

Parameter	Symbol	Condition	Macronix MX29GL128F	Micron 28F128M29EW
VCC Rise Time	Tvr	min	20us/V	N/A
		max	500ms/V	N/A
VCC Setup Time	Tvcs/tVCHPH+tPHEL	min	500us	60us
VIO Setup Time	Tvios or tVIOS	min	N/A	50us

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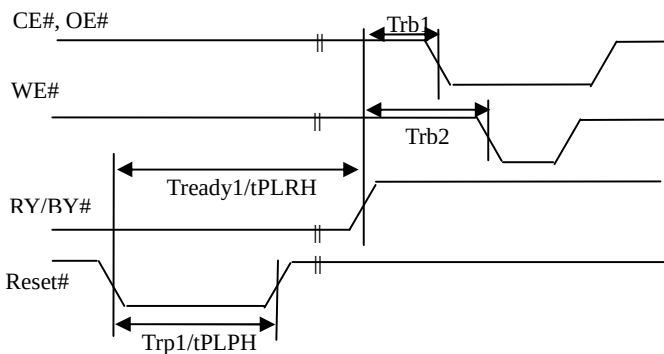
9. H/W Reset# Pin Timing

Macronix and Micron both offer a hardware reset function. The operation is same but the timing is slightly different. System timing may need to be adjusted for the auto algorithm and non-auto algorithm (Figure 9-1).

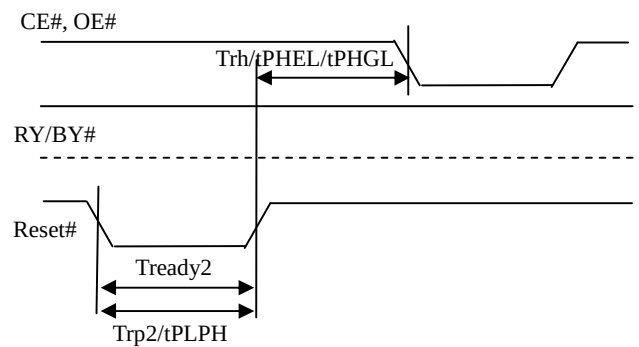
Table 9-1: Reset# Timing

H/W Timing Characteristic		Macronix MX29GL128F	Micron 28F128M29EW
Trh or tPHEL/tPHGL (Reset# high to Read)		200ns (min)	50ns (min)
During auto algorithm mode *	Trp1 or tPLPH (Reset# pulse width)	10us (min)	100ns (min)
	Tready1 or tPLRH (Reset# low to read or write)	20us (max)	25us (max)
During non-auto algorithm mode *	Trp2 or tPLPH (Reset# pulse width)	500ns (min)	100ns (min)
	Tready2 (Reset# low to read or write)	500ns (max)	N/A

* Macronix defines different reset timing between auto and non-auto algorithm; Micron defines same value in both.



During auto algorithm mode



During non-auto algorithm mode

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10. Summary

Macronix MX29GL128F and Micron® 28F128M29EW Parallel Flash have similar commands, functions, and features. Additionally, the supported package types have identical footprints and are pin compatible. The software may need to be modified to use a maximum Write Buffer size of 32-Words. In addition, the software algorithms need to ensure that Write Buffer addresses fall within a 32-Word page boundary. There are access time differences and accelerated programming voltage level differences which may also need to be accommodated.

11. Data Sheet Version

Table 11-1 shows the datasheet versions used for comparison in this application note. For the most current, detailed Macronix specification, please refer to the Macronix Website at <http://www.macronix.com>

Table 11-1: Datasheet Version

Data sheet	Location	Date Issue	Revision
MX29GL128F	Website	Jun. 28, 2012	Rev. 1.1
M29EW_32mb_128mb	Website	Aug 2012	Rev. A

12. Appendix

Cross Reference Table 12-1 shows basic part number and package information for the Macronix MX29GL128F and Micron® 28F128M29EW products.

Table 12-1: Part Number Cross Reference

Density	Macronix Part	Micron Part	Package	Dimension
128Mb	MX29GL128FUT2I-11G	JS28F128M29EWHF	56-TSOP (lead-free)	14x20mm
	MX29GL128FDT2I-11G	JS28F128M29EWLA		
	MX29GL128FUXFI-11G	PC28F128M29EWHF	64-LFBGA (lead-free)	11x13x14mm 0.6mm ball
	MX29GL128FDXFI-11G	PC28F128M29EWLA		
	MX29GL128FUXFI-11G*	RC28F128M29EWHF	64-LFBGA (leaded)	11x13x14mm 0.6mm ball
	MX29GL128FDXFI-11G*	RC28F128M29EWLA		

Note: * = Macronix does not offer parts with lead.

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